CONSTRAINED RAMSEY NUMBERS FOR THE LOOSE 3-UNIFORM PATH

Abstract. Let $\mathcal{L}$ be the loose 3-uniform path of length three. We prove that

$$f(H,\mathcal{L}) = \max\{R_2(H), 7\}$$

for every 3-graph H with at least two edges. The new case is the order boundary $R_2(H) = |V(H)| \geq 7$, and therefore the result answers Problem 7.2 of Li and Wang. The proof uses their structural theorem for rainbow-$\mathcal{L}$-free colorings, their transfer argument for the second structural case, and a necessary structural condition on 3-graphs attaining the order bound for their two-color Ramsey number.

1. Introduction

Let

$$\mathcal{L} = \{\{v_1,v_2,v_3\},\{v_3,v_4,v_5\},\{v_5,v_6,v_7\}\}$$

be the loose 3-uniform path of length three. For a 3-graph H, let $R_2(H)$ be the least n such that every two-coloring of $K_n^{(3)}$ contains a monochromatic copy of H. For 3-graphs H and G, let $f(H,G)$ be the least n such that every edge-coloring of $K_n^{(3)}$, with an arbitrary number of colors, contains either a monochromatic H or a rainbow G. Every copy of H in $K_n^{(3)}$ occupies $|V(H)|$ vertices, so $K_n^{(3)}$ contains no copy of H at all when $n < |V(H)|$; hence we have the *order bound*

$$R_2(H) \geq |V(H)|.$$

This inequality is used twice in the proof of Theorem 1.

Liu proved $f(H,\mathcal{L}) = R_2(H)$ when $R_2(H) \geq \max\{|V(H)|+3, 10\}$ [1]. Li and Wang improved this to

$$R_2(H) \geq \max\{|V(H)|+1, 7\}$$

and asked whether equality still holds at the remaining boundary $R_2(H) = |V(H)| \geq 7$ [2, Theorem 1.10 and Problem 7.2]. We answer their problem affirmatively and obtain the complete formula.

Theorem 1. *For every 3-graph H with at least two edges,*

$$f(H,\mathcal{L}) = \max\{R_2(H), 7\}.$$

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For a 3-graph with one edge, one trivially has $f(H,\mathcal{L}) = R_2(H) = |V(H)|$. Thus Theorem 1 determines $f(H,\mathcal{L})$ for every nonempty 3-graph H.

We use the following theorem of Li and Wang [2, Theorem 1.5].

Theorem 2 (Li–Wang)**.** *Let $n \ge 7$, and let c be an edge-coloring of $K_n^{(3)}$ using at least three colors and containing no rainbow $\mathcal{L}$. Then at least one of the following holds:*

1. *there is a vertex u such that $K_n^{(3)} - u$ is monochromatic;*
2. *there are an edge e and a color $\alpha \ne c(e)$ such that every edge $f \ne e$ with $c(f) \ne \alpha$ satisfies $|e \cap f| = 2$.*

2. The order boundary

A 3-graph H is a *cone* over a graph J if

$$V(H) = \{x\} \cup V(J), \qquad E(H) = \big\{\{x,p,q\} : pq \in E(J)\big\}.$$

The vertex x is the cone vertex and J is its link graph. We write $K_a + K_b$ for the disjoint union of K_a and K_b.

Lemma 3 (Boundary structure)**.** *Let H be a 3-graph on $n \ge 7$ vertices with $R_2(H) = n$. Then either H has an isolated vertex or H is a cone over a graph J with $\delta(J) = 1$.*

Proof. Fix $z \in V(K_n^{(3)})$, and color a triple red if it contains z and blue otherwise. Since every copy of H in $K_n^{(3)}$ is spanning, a red copy makes H a cone and a blue copy gives an isolated vertex. Thus, if H has no isolated vertex, it is a cone over a graph J on $m = n-1$ vertices, and $\delta(J) \ge 1$.

Suppose that $\delta(J) \ge 2$. For $A \subseteq V(K_n^{(3)})$, color each triple T by $|T \cap A| \pmod 2$. If $|A| = a$, then the two links at a vertex are

$$\begin{cases} K_{a-1,n-a} \text{ and } K_{a-1} + K_{n-a}, & y \in A, \\ K_{a,n-a-1} \text{ and } K_a + K_{n-a-1}, & y \notin A. \end{cases}$$

A monochromatic copy of the cone would place a spanning copy of J in one of these links.

First assume $n \ge 8$, so $m \ge 7$. With $a = 2$, either the coloring avoids H, or J is a spanning subgraph of one of

$$K_{1,m-1}, \quad K_1 + K_{m-1}, \quad K_{2,m-2}, \quad K_2 + K_{m-2}.$$

The minimum-degree condition excludes all but $K_{2,m-2}$ and then forces $J \cong K_{2,m-2}$. Taking $a = 4$, the possible links are

$$K_{3,m-3}, \quad K_3 + K_{m-3}, \quad K_{4,m-4}, \quad K_4 + K_{m-4}.$$

None contains a spanning $K_{2,m-2}$: the disconnected links cannot contain a connected spanning graph, while a connected bipartite graph has a unique bipartition and the complete bipartite links have different part sizes. Hence one of the two parity colorings avoids H, contradicting $R_2(H) = n$.

It remains to treat $n = 7$. The $a = 2$ coloring either avoids H or forces $J \cong K_{2,4}$. In the latter case, work on $\mathbb{Z}_7$ and set

$$D_1 = \{0,1,3\}, \qquad D_2 = \{0,2,3\}, \qquad \mathcal{B}_i = \{t + D_i : t \in \mathbb{Z}_7\}.$$

The ordered nonzero differences of each D_i are all six nonzero residues, so each $\mathcal{B}_i$ is a Steiner triple system. The two systems are disjoint: the translates of D_1 containing 0 are $\{0,1,3\}$, $\{0,2,6\}$, and $\{0,4,5\}$, none of which equals D_2. Color the fourteen blocks in $\mathcal{B}_1 \cup \mathcal{B}_2$ red and all other triples blue. At every vertex, the red link is the union of two edge-disjoint perfect matchings, hence a 6-cycle; the blue link is its 3-regular complement in K_6. Neither link contains a spanning $K_{2,4}$: the red link has six edges and the blue link has maximum degree three. This again contradicts $R_2(H) = 7$.

Therefore $\delta(J) = 1$. □

The next lemma is not new: it is the argument Li and Wang give for case (ii), which is part (2) above, in their proof of [2, Theorem 1.10], where the recoloring is set up through the two isomorphic spanning subgraphs $F_1' = \{v_iv_jv_\ell : i,j \in [3],\ \ell \notin [3]\} \cup \{v_1v_2v_3\}$ and F_2' defined in the same way from $\{v_4, v_5, v_6\}$; the permutation π below is their isomorphism $F_1' \to F_2'$. Their proof is stated for $n = R_2(H)$, and it extends without change to $n \geq R_2(H)$, which is the only form we need. We include it for completeness and because the proof of Theorem 1 applies it at $N = \max\{R_2(H), 7\}$.

Lemma 4 (Transfer; Li and Wang [2])**.** *Let $n \geq 7$, let $R_2(H) \leq n$, and suppose that a rainbow-$\mathcal{L}$-free coloring of $K_n^{(3)}$ satisfies part (2) of Theorem 2. Then it contains a monochromatic copy of H.*

Proof. Relabel so that $e = \{1,2,3\}$, choose distinct vertices $4, 5, 6$ outside e, and let

$$\pi = (1\ 4)(2\ 5)(3\ 6)$$

fix all remaining vertices. Let S be the set of edges whose color is not α. If $f = e$, then $\pi(f)$ is disjoint from e; if $f \neq e$, then $|f \cap e| = 2$ and $|\pi(f) \cap e| \leq 1$. In either case $\pi(f) \neq e$, so the contrapositive of part (2) gives $c(\pi(f)) = \alpha$.

Recolor the edges in S blue and all other edges red. Since $R_2(H) \leq n$, this two-coloring contains a monochromatic H. A red copy is already α-monochromatic in the original coloring, while the image under π of a blue copy is α-monochromatic. □

3. Proof of the formula

Proof of Theorem 1. Write $r = R_2(H)$ and $N = \max\{r, 7\}$. A two-coloring of $K_{r-1}^{(3)}$ without a monochromatic H contains no rainbow $\mathcal{L}$, so $f(H, \mathcal{L}) \geq r$. Also, coloring all triples of $K_6^{(3)}$ differently gives neither a monochromatic H nor a rainbow $\mathcal{L}$. Hence $f(H, \mathcal{L}) \geq N$.

For the reverse inequality, let c be a rainbow-$\mathcal{L}$-free coloring of $K_N^{(3)}$. If at most two colors occur, then c contains a monochromatic H because $N \geq r$. Assume that at least three colors occur and apply Theorem 2.

If part (2) holds, Lemma 4 gives a monochromatic H. Suppose part (1) holds, and let u be such that all triples in $W = V(K_N^{(3)}) \setminus \{u\}$ have one color, say α.

If $r \le 6$, then $N = 7$ and, by the order bound, $|V(H)| \le r \le 6$, so the monochromatic $K_6^{(3)}$ on W contains H. Now let $r \ge 7$, so $N = r$. If $r \ge |V(H)| + 1$, the same conclusion is immediate; this is the step Li and Wang use for case (i), that is part (1), in their proof of [2, Theorem 1.10]. By the order bound it remains only to consider

$$r = |V(H)| \ge 7.$$

By Lemma 3, either H has an isolated vertex or H is a cone over a graph J with $\delta(J) = 1$. In the first case, map an isolated vertex to u and all remaining vertices bijectively to W. This gives an α-monochromatic copy of H.

In the second case, let x be the cone vertex, choose a leaf p of J, and let q be its unique neighbor. If $c(\{u, y, w\}) = \alpha$ for some distinct $y, w \in W$, map x, p, q to y, u, w, respectively, and extend bijectively. The edge corresponding to pq maps to $\{u, y, w\}$, and every other edge maps into W; hence the resulting copy of H is α-monochromatic.

We may therefore assume that every triple through u has color different from α. Color each pair $ab \in \binom{W}{2}$ by $c(\{u, a, b\})$. If ab and cd are disjoint, choose distinct $r', s' \in W \setminus \{a, b, c, d\}$. Then

$$\{u, a, b\}, \qquad \{u, c, d\}, \qquad \{c, r', s'\}$$

form a copy of $\mathcal{L}$, and the last edge has color α. The first two edges must therefore have the same color. The graph on $\binom{W}{2}$ joining disjoint pairs is connected for $|W| \ge 6$: two distinct intersecting pairs have a common disjoint pair. Thus all triples through u have one common color, different from α. Hence c uses only two colors, a contradiction.

Every rainbow-$\mathcal{L}$-free coloring of $K_N^{(3)}$ contains a monochromatic H. Hence $f(H, \mathcal{L}) \le N$, completing the proof. $\square$

The case $R_2(H) = |V(H)| \ge 7$ in the proof gives, in particular, an affirmative answer to Problem 7.2 of Li and Wang.

Status and scope

What is settled is Problem 7.2 of [2] as stated there: for every 3-graph H with $R_2(H) = |V(H)| \ge 7$ one has $f(H, \mathcal{L}) = R_2(H)$. Problem 7.2 imposes no restriction on the number of edges of H; Theorem 1 assumes $|E(H)| \ge 2$, and the one-edge case is disposed of in the paragraph following it, so the problem is answered in full.

Two of the ingredients are due to Li and Wang. Theorem 2 is [2, Theorem 1.5], and Lemma 4 is the case-(ii) argument in their proof of [2, Theorem 1.10], in different notation and with $n = R_2(H)$ relaxed to $n \ge R_2(H)$; the step that disposes of $r \ge |V(H)| + 1$ in case (1) is likewise theirs. We have not been able to consult the full text of [1], so we do not know whether the recoloring-and-transfer device of Lemma 4 originates with [2] or already with Liu; in either case it is not new here. What is new is Lemma 3 together with the treatment of case (1) at $r = |V(H)|$, which is the case Problem 7.2 asks about.

Lemma 3 states a necessary condition and nothing more. It does not classify the 3-graphs with $R_2(H) = |V(H)| \geq 7$, and no converse is proved or claimed: a 3-graph with an isolated vertex, or a cone over a graph of minimum degree 1, need not attain the order bound.

The coloring used in the proof of Lemma 3 for $n = 7$ is a pair of disjoint Steiner triple systems on $\mathbb{Z}_7$, that is, a pair of disjoint Fano planes. Such a pair is classical and is not claimed here as a new object; what is used is only that the two links it produces at a vertex, a 6-cycle and its 3-regular complement in K_6, both fail to contain a spanning $K_{2,4}$, which is what rules out a monochromatic cone over $K_{2,4}$ in $K_7^{(3)}$.

Exact verification. Every verification above is finite and hand-sized, and each one is carried out from the data printed in this paper: the six ordered nonzero differences of each of $D_1 = \{0, 1, 3\}$ and $D_2 = \{0, 2, 3\}$, which give the Steiner property of $\mathcal{B}_1$ and $\mathcal{B}_2$; the three translates of D_1 containing 0, which give their disjointness; and the two links at a vertex of the resulting coloring of $K_7^{(3)}$, which give the failure of a spanning $K_{2,4}$. All seventeen of these checks were recomputed a second time, independently and from the printed residues alone, and every one agreed. No program is needed and none accompanies this paper; a reader can redo every step from the paper by hand.

We have found no answer to Problem 7.2 in the literature: the only paper we found citing [2] is the companion paper [3], on graphs and rainbow P_5, which does not address $f(H, \mathcal{L})$; we found no journal version of [2], and its latest arXiv revision still records Problem 7.2 as open. Our search is not exhaustive, and the result is offered subject to external prior-art review.

References

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